「Section 5.1 If $f: \mathbb{R} \to \mathbb{R}$ satisfies that

$$|f(x)-f(y)| \le (x-y)^2 \quad \text{for all } x, y \in \mathbb{R}$$

Then f is constant.

Proof. Since $\left|\dfrac{f(x)-f(y)}{x-y}\right| \le |x-y|$ for all $x, y \in \mathbb{R}$,

given $x \in \mathbb{R}$, $f'(x)$ exists with $f'(x)=0$.

Now, by the Mean value Theorem, for any $a<b$, there exists $c \in (a,b)$ s.t

$$f(a)-f(b) = (a-b)\cdot f'(c) = 0$$

Thus f is constant.

「Section 5.2. Let $f: (a,b) \to \mathbb{R}$ be given. If $f'(x) > 0$ for all $x \in (a,b)$, then f is injective. Moreover, the inverse $f^{-1}: f[(a,b)] \to (a,b)$ exists and differentiable with

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad (a<x<b)$$

Proof. Let $a<x<y<b$. Then, by the Mean-Value Theorem, there exists $c \in (x,y)$ s.t

$$f(y)-f(x) = f'(c)\cdot(y-x) > 0$$

Therefore, f is strictly increasing, thus injective. Now, the restriction

$$f: (a,b) \to f[(a,b)]$$

is bijection, thus there exists a inverse

$$f^{-1}: f[(a,b)] \to (a,b) : f(x) \mapsto x$$

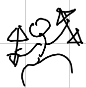
Now, $(f^{-1} \circ f)(x) = x$ implies that

Section 5.3. Let $g : \mathbb{R} \to \mathbb{R}$ be given with $|g'| \le M$ for some $M > 0$.
Fix $\varepsilon > 0$, and define $f(x) = x + \varepsilon \cdot g(x)$. prove that f is injective if ε is small enough.

Proof. If $\varepsilon < \frac{1}{M}$, then $f'(x) = 1 + \varepsilon \cdot g'(x) > 0$. Therefore f is injective.

Section 5.4. If $C_0, \cdots, C_n \in \mathbb{R}$ satisfies

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$$



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, then the polynomial

$$C_0 + C_1 x + C_2 x^2 + \cdots + C_{n-1} x^{n-1} + C_n x^n = p(x)$$

has a real root in $(0, 1)$.

Proof. Define $g(x) = C_0 x + \frac{C_1}{2} x^2 + \cdots + \frac{C_{n-1}}{n} x^n + \frac{C_n}{n+1} x^{n+1}$

Then, $g(0) = 0$ clearly, and $g(1) = 0$ by the assumption.
By the Mean-Value Theorem, there exists $\alpha \in (0, 1)$ s.t

$$0 = g(1) - g(0) = g'(\alpha) \cdot (1 - 0) = g'(\alpha)$$

Directly, $p(x) = g'(x)$. therefore $g'(\alpha) = p(\alpha) = 0$.

Section 5.5 Let $f : (0, \infty) \to \mathbb{R}$ be differentiable.
If $f'(x) \to 0$ as $x \to \infty$, then $f(x+1) - f(x) \to 0$ $x \to \infty$.

Proof. By the MVT, for any $x > 0$, there exists $C_x \in (x, x+1)$ s.t

$$f(x+1) - f(x) = f'(C_x) \cdot (1 - 0) = f'(C_x).$$

Let $\varepsilon > 0$ be given. Then, there exists $\delta > 0$ s.t $x \ge \delta \Rightarrow |f'(x)| < \varepsilon$.
For this $\delta > 0$, for any $x \ge \delta$, there is $C_x \in (x, x+1)$ s.t

$$|f(x+1) - f(x)| = |f'(C_x)| < \varepsilon$$

Thus proved.

「Section 5.6. Suppose that $f:(0,\infty)\to\mathbb{R}$ satisfies that;

1). f is continuous for $x\geq 0$
2). f' exists for $x>0$
3). f' increases monotonically.
4). $f(0)=0$.

Then, $g:(0,\infty)\to\mathbb{R}: x\mapsto \frac{f(x)}{x}$ increases monotonically.

Proof. Since $g'(x)=\dfrac{xf'(x)-f(x)}{x^2}$, enough to show $f'(x)\geq \frac{f(x)}{x}=g(x)$.

observe that $g(x)=\dfrac{f(x)}{x}=\dfrac{f(x)-f(0)}{x-0}=f'(c_x)$ for some $0<c_x<x$, by the MVT.

Now, since f' increases monotonically $f'(x)\geq g(x)=f'(c_x)$.

「Section 5.7. Let $f,g:\mathbb{R}\to\mathbb{R}$ be given.

Suppose that f', g' exist and $g'(x)\neq 0$ for all $x\in\mathbb{R}$, and $f(x)=g(x)=0$.

Prove $\displaystyle\lim_{t\to x}\frac{f(t)}{g(t)}=\frac{f'(t)}{g'(t)}$

Proof. Since f', g' exist, and $f(x)=g(x)=0$,

$$\lim_{t\to x}\frac{f(t)}{g(t)}=\lim_{t\to x}\frac{\dfrac{f(t)-f(x)}{t-x}}{\dfrac{g(t)-g(x)}{t-x}}=\frac{f'(x)}{g'(x)}$$

「 $\displaystyle\sum_{n=1}^{\infty}\sin\left(\frac{1}{n}\right)$ Converges?

$\dfrac{\sin\frac{1}{n}}{\frac{1}{n}}\to 1$ but $\sum\frac{1}{n}$ diverges, thus

「Section 5. 8. Let $f: [a,b] \to \mathbb{R}$ be given. Suppose f' exists and Continuous on $[a,b]$.
Given $\varepsilon > 0$, there exists $\delta > 0$ s.t. $\forall x, t \in [a,b]$, $|x-t| < \delta \Rightarrow \left| \frac{f(x) - f(t)}{x - t} - f'(x) \right| < \varepsilon$

Proof. Since f' is Continuous on Compact, it is Continuous Uniformly.
Given $\varepsilon > 0$, there exists $\delta > 0$ s.t. $\forall x, t \in [a,b]$,

$$|x-t| < \delta \Rightarrow \left| f'(x) - f'(t) \right| < \varepsilon$$

Meanwhile, by the Mean-Value Thm,

$$f'(s) = \frac{f(x) - f(t)}{x - t}$$ for some s between x and t.

Now, $|x-t| < \delta \Rightarrow \left| \frac{f(x) - f(t)}{x - t} - f'(t) \right| = \left| f'(s) - f'(t) \right| < \varepsilon$

because $|s-t| \leq |x-t|$.

「Section 5.9.

$$f: \mathbb{R} \to \mathbb{R} ; x \mapsto \begin{cases} x+3 & x \geq 0. \\ 3 & x < 0 \end{cases}$$

「Section 5.10. Let $f, g : (0,1) \to \mathbb{C}$ be differentiable such that $\lim_{x \to 0} f(x) = \lim_{x \to 0} g(x) = 0$ and $\lim_{x \to 0} f'(x)$, $\lim_{x \to 0} g'(x) (\neq 0)$ exist. Then,

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = \lim_{x \to 0} \frac{f'(x)}{g'(x)}$$

Proof. observe that $\frac{f(x)}{g(x)} = \left[\frac{f(x)}{x} - \lim_{t \to 0} f'(t) \right] \cdot \frac{x}{g(x)} + \lim_{t \to 0} f'(t) \cdot \frac{x}{g(x)}$

Meanwhile, by the L'Hopital Law,

$$\lim_{x \to 0} \frac{f(x)}{x} = \lim_{x \to 0} \left[\frac{\operatorname{Re} f(x)}{x} + i \cdot \frac{\operatorname{Im} f(x)}{x} \right]$$

$$= \lim_{x \to 0} \left[\operatorname{Re} f'(x) + i \cdot \operatorname{Im} f'(x) \right] = \lim_{x \to 0} f'(x)$$

Similarly $\lim_{x \to 0} \frac{x}{g(x)} = \left(\lim_{x \to 0} g'(x) \right)^{-1}$. Now,

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = \left(\lim_{x \to 0} f'(x) \right) \cdot \left(\lim_{x \to 0} g'(x) \right)^{-1}$$

「Section 5.11.

Suppose that f is defined on a neiborhood of $x \in \mathbb{R}$, and $f''(x)$ exists. Prove that

$$\lim_{h \to 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x)$$

Proof. By the L'Hopital Law,

$$\lim_{h \to 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} \overset{①}{=} \lim_{h \to 0} \frac{f'(x+h) - f'(x-h)}{2h} \overset{②}{=} \frac{1}{2} \cdot \lim_{h \to 0} \left[\frac{f'(x+h) - f'(x)}{h} + \frac{f'(x-h) - f'(x)}{-h} \right]$$

$$= \frac{1}{2} \cdot 2f''(x).$$

Define $f : \mathbb{R} \to \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$

Then, $f''(0)$ does not exists, But

$$\lim_{h \to 0} \frac{h^2 + 0 - 0}{h^2} = 1$$

exists.

$$\frac{1}{2} \cdot \lim_{h \to 0} \left(f''(x+h) + f''(x-h) \right) = f''(x)$$

「Fail Try

By the Mean-Value Theorem, for any $h > 0$,

$f(x+h) - f(x) = f'(t_1) h$ and $f(x-h) - f(x) = f'(t_2) h$ for some

for some $t_1 \in (x, x+h)$, $t_2 \in (x-h, x)$

$$\frac{f'(t_1) + f'(t_2)}{h} = 2f''(x)$$

[Section 5.13, Let $a \in \mathbb{R}$, $c > 0$. Define

$$f : [-1,1] \to \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x = 0 \\ x^a \cdot \sin\left(\frac{1}{|x|^c}\right) & \text{if } x \neq 0 \end{cases}$$

Equivalent Condition Table	f	f'	f''
Continuous	$a > 0$	$a > 1+c$	$a > 2+2c$
bounded	$a \geq 0$	$a \geq 1+c$	$a \geq 2+2c$
Exists at $x=0$	Always...	$a > 1$	$a > 2+c$

[Proof. Suppose that $a > 0$. Given $\varepsilon > 0$, put $\delta = \sqrt[a]{\varepsilon}$. Then,

$$|x| < \delta \Rightarrow \left|x^a \cdot \sin\left(\frac{1}{|x|^c}\right)\right| \leq |x|^a < \varepsilon.$$

Thus continuous. And, suppose that $a \leq 0$. Then, a sequence

$$x_n = \sqrt[c]{\frac{1}{(2n+\frac{1}{2})\pi}}$$

converges to 0, but

$$f(x_n) = \left(\frac{1}{(2n+\frac{1}{2})\pi}\right)^{\frac{a}{c}} \cdot \underbrace{\sin\left(2n\pi + \frac{\pi}{2}\right)}_{=1}$$

not converges to $f(0) = 0$. Except $x \neq 0$, clearly continuous,

$f'(0)$ exists iff

$$\lim_{t \to 0} \frac{f(t) - f(0)}{t - 0} = \lim_{t \to 0}\left(t^{a-1} \cdot \sin\left(\frac{1}{|t|^c}\right)\right)$$

exists. Since it equals to $t^{-1} \cdot f(t)$, the limit exists only $a - 1 > 0$.

Suppose $a \geq 1+c$. Then, for $x \neq 0$,

Observe that: for $x > 0$,

$$f'(x) = ax^{a-1}\sin\left(\frac{1}{x^c}\right) + x^a \cdot (-c) \cdot x^{-c-1} \cdot \cos\left(\frac{1}{x^c}\right)$$

$$= ax^{a-1}\sin\frac{1}{x^c} - c \cdot x^{a-(1+c)} \cdot \cos\frac{1}{x^c}$$

is bounded iff $a - (1+c) \geq 0$.

d). f' is continuous at $x=0$ iff $f'(0) = \lim_{x\to 0} f'(x)$ (clearly, existence of $f'(0)$ is assumed.
If $a > 1+c$, then $f(x) \to 0$ as $x \to 0$, thus

$$f'(0) = \lim_{x\to 0} \frac{f(x)}{x} = \lim_{x\to 0} f'(x)$$

by the L'Hopital Law. And, suppose $a \le 1+c$. Then,

$$f'(x) = a x^{a-1} \sin \frac{1}{x^c} - c \cdot x^{a-(1+c)} \cdot \cos \frac{1}{x^c}$$

does not converges as $x \to 0$.

「Section 5, 14. Let $f : (a,b) \to \mathbb{R}$ be differentiable.

f is convex if and only if f' is monotonically increasing.

「Section 5.15. Let $f:(a,\infty)\to\mathbb{R}$ be twice-differentiable.
Put $M_k=\sup_{x\in(a,\infty)}|f^{(k)}(x)|$, $k=0,1,2$. Then $M_1^2\le 4M_0M_2$.

proof. Given $x>a$ and $h>0$,

$$f(x+2h)=f(x)+f'(x)\cdot 2h+f''(t)\cdot\underbrace{2h^2}_{=\frac{4h^2}{2!}} \quad \text{for some } t\in(x,x+2h)$$

Therefore, $|f'(x)|\le\left|\frac{f(x+2h)-f(x)}{2h}-h\cdot f''(t)\right|\le\frac{M_0}{h}+M_2h$

Take $h=\sqrt{\frac{M_0}{M_2}}$. Then,

$$\forall x>a,\ |f'(x)|\le\sqrt{M_0M_2}+\sqrt{M_0M_2}=2\sqrt{M_0M_2}$$

which implies that $M_1\le 2\sqrt{M_0M_2}$.

「Section 5.16. Let $f:(0,\infty)\to\mathbb{R}$ be twice-differentiable,
f'' is bounded,
$f(x)\to 0$ as $x\to\infty$.

Then, $f'(x)\to 0$ as $x\to\infty$.

Proof. Since for any $a>0$,

$$0\le\sup_{x\in(a,\infty)}|f'(x)|\le 2\cdot\sqrt{\sup_{x\in(a,\infty)}|f(x)|\cdot M} \quad (\text{where } M \text{ is bound for } |f''|)$$

Letting $a\to\infty$, we obtain the result.

「Section 5.17. Let $f:[-1,1]\to\mathbb{R}$ be three-times differentiable s.t

$$f(-1)=0,\ f(0)=0,\ f(1)=1,\ f'(0)=0.$$

Then, $f^{(3)}(x)\geq 3$ for some $x\in(-1,1)$.

Proof. By the Taylor's Theorem,

$$f(1)=f(0)+f'(0)+\frac{f''(0)}{2!}+\frac{f^{(3)}(s)}{3!} \quad \text{for some } s\in(0,1)$$

and
$$f(-1)=f(0)-f'(0)+\frac{f''(0)}{2!}-\frac{f^{(3)}(t)}{3!} \quad \text{for some } t\in(-1,0)$$

Combining above, then

$$f(1)-f(-1)=2f'(0)+\frac{1}{3!}\left(f^{(3)}(s)+f^{(3)}(t)\right)$$

Thus $f^{(3)}(s)+f^{(3)}(t)=6$, therefore either one bigger than 3.

Q. Why $f(0)=0$ is needed?

「Section 5.18.

Let $f:[a,b]\to\mathbb{R}$, $n\in\mathbb{N}$, $\alpha,\beta\in[a,b]$. Suppose $f^{(n-1)}$ exists on $[a,b]$. Define

$$Q(t)=\frac{f(t)-f(\beta)}{t-\beta} \quad (t\neq\beta).$$

Then
$$f(\beta)=\sum_{k=0}^{n-1}\frac{f^{(k)}(\alpha)}{k!}\cdot(\beta-\alpha)^k+\frac{Q^{(n-1)}(\alpha)}{(n-1)!}\cdot(\beta-\alpha)^n$$

Proof. Derivating $(n-1)$ times to:

$$(t-\beta)\cdot Q(t)=f(t)-f(\beta)$$

Then,
$$Q(t)+(t-\beta)Q'(t)=f'(t)$$
$$Q'(t)+Q'(t)+(t-\beta)Q''(t)=f''(t)$$
$$\vdots$$
$$(n-1)\cdot Q^{(n-2)}(t)+(t-\beta)\cdot Q^{(n-1)}(t)=f^{(n-1)}(t)$$

Now,
$$\begin{aligned}
f(\beta)&=f(t)+Q(t)\cdot(\beta-t)\\
&=f(t)+f'(t)\cdot(\beta-t)+Q'(t)(\beta-t)^2\\
&=f(t)+f'(t)(\beta-t)+\frac{f''(t)}{2!}(\beta-t)^2+\frac{Q''(t)}{2!}(\beta-t)^3\\
&\ \ \vdots\\
&=\sum_{k=0}^{n-1}\frac{f^{(k)}(t)}{k!}\cdot(\beta-t)^k+\frac{Q^{(n-1)}(t)}{(n-1)!}\cdot(\beta-t)^n.
\end{aligned}$$

「Section 5. 19. Let $f:(-1,1)\to\mathbb{R}$ be a function such that $f'(0)$ exists.
Suppose $-1<\alpha_n<\beta_n<1$ and $\lim_{n\to\infty}\alpha_n=\lim_{n\to\infty}\beta_n=0$. Define the difference quotient:

$$D_n=\frac{f(\beta_n)-f(\alpha_n)}{\beta_n-\alpha_n}$$

1). If $\alpha_n<0<\beta_n$, then $\lim_{n\to\infty}D_n=f'(0)$

2). If $0<\alpha_n<\beta_n$ and $\left\{\frac{\beta_n}{\beta_n-\alpha_n}\right\}$ is bounded, then $\lim_{n\to\infty}D_n=f'(0)$

3). If f' is continuous in $(-1,1)$, then $\lim_{n\to\infty}D_n=f'(0)$

4). Find Counter examples.

Proof. Observe that

$$\left|\frac{f(\beta_n)-f(\alpha_n)}{\beta_n-\alpha_n}-f'(0)\right|=\left|\frac{f(\beta_n)-f(0)}{\beta_n-\alpha_n}-\frac{f(\alpha_n)-f(0)}{\beta_n-\alpha_n}-f'(0)\right|$$

$$=\left|\frac{f(\beta_n)-f(0)}{\beta_n-0}\cdot\frac{\beta_n}{\beta_n-\alpha_n}-\frac{\beta_n}{\beta_n-\alpha_n}\cdot f'(0)-\frac{f(\alpha_n)-f(0)}{\alpha_n-0}\cdot\frac{\alpha_n}{\beta_n-\alpha_n}-\frac{\alpha_n}{\beta_n-\alpha_n}\cdot f'(0)\right|$$

$$\le\left|\frac{\beta_n}{\beta_n-\alpha_n}\right|\cdot\left|\frac{f(\beta_n)-f(0)}{\beta_n-0}-f'(0)\right|+\left|\frac{\alpha_n}{\beta_n-\alpha_n}\right|\cdot\left|\frac{f(\alpha_n)-f(0)}{\alpha_n-0}-f'(0)\right|\quad(*)$$

Now, if $\left\{\frac{\beta_n}{\beta_n-\alpha_n}\right\}$ is bounded, $(*)$ tends to 0 as $n\to\infty$. $\left(\left|\frac{\alpha_n}{\beta_n-\alpha_n}\right|\le\left|\frac{\beta_n}{\beta_n-\alpha_n}\right|\right)$.

If $\alpha_n<0<\beta_n$, $\left|\frac{\beta_n}{\beta_n-\alpha_n}\right|\le\left|\frac{\beta_n}{\beta_n}\right|=1$ and $\left|\frac{\alpha_n}{\beta_n-\alpha_n}\right|\le\left|\frac{\alpha_n}{\alpha_n}\right|=1$

Thus 1), 2) are proved.

And 3) can be solved by applying MVT. For any $n\in\mathbb{N}$, there exists $c_n\in(\alpha_n,\beta_n)$ s.t.

$$D_n=\frac{f(\beta_n)-f(\alpha_n)}{\beta_n-\alpha_n}=f'(c_n)$$

Now, $c_n\to0$ as $n\to\infty$ and f' is continuous,

$$D_n=f'(c_n)\to f'(0)\quad\text{as } n\to\infty.$$

「Try for 18.

$$= \left|\frac{\beta_n}{\beta_n - \alpha_n}\right| \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} - \frac{f(\alpha_n) - f(0)}{\beta_n} - \frac{(\beta_n - \alpha_n)\cdot f'(0)}{\beta_n} \right|$$

$$= \left|\frac{\beta_n}{\beta_n - \alpha_n}\right| \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} - f'(0) - \left(\frac{f(\alpha_n) - f(0)}{\beta_n} - \frac{\alpha_n}{\beta_n}\cdot f'(0) \right) \right|$$

$$\le \left|\frac{\beta_n}{\beta_n - \alpha_n}\right| \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} - f'(0) \right| + \left|\frac{\alpha_n}{\beta_n - \alpha_n}\right| \cdot \left| \frac{f(\alpha_n) - f(0)}{\alpha_n - 0} - f'(0) \right|$$

$$\left| \frac{f(\beta_n) - f(0)}{\beta_n - \alpha_n} \right| = \left| \frac{\beta_n}{\beta_n - \alpha_n} \cdot \frac{f(\beta_n) - f(0)}{\beta_n - 0} \right| \le M \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} \right| \to M \cdot |f'(0)|.$$

Given $\varepsilon > 0$, there exists $\delta > 0$ s.t $0 < |x| < \delta \Rightarrow \left| \frac{f(t) - f(0)}{t} - f'(0) \right| < \varepsilon$.
, since $f'(0)$ exists. Take $N \in \mathbb{N}$ s.t

$$n \ge N \Rightarrow -\frac{\delta}{2} < \alpha_n < 0 \quad \text{and} \quad 0 < \beta_n < \frac{\delta}{2}$$

(Note: 여기서 이상함을 느끼게 되는데, 차분 형태라 바로 적용은 불가능 어떻게 할까?)

Meanwhile, since $f'(0)$ exists, f is continuous at $x=0$. Therefore,

$$\lim_{n\to\infty} f(\alpha_n) = \lim_{n\to\infty} f(\beta_n) = f(0)$$

$$\left| \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} - f'(0) \right| = \left| \frac{f(\beta_n) - f(0)}{\beta_n - \alpha_n} - \frac{f(\alpha_n) - f(0)}{\beta_n - \alpha_n} - f'(0) \right|$$

$$= \left|\frac{\beta_n}{\beta_n - \alpha_n}\right| \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} - \frac{f(\alpha_n) - f(0)}{\beta_n} - \frac{(\beta_n - \alpha_n)\cdot f'(0)}{\beta_n} \right|$$

$$= \left|\frac{\beta_n}{\beta_n - \alpha_n}\right| \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} - f'(0) - \left(\frac{f(\alpha_n) - f(0)}{\beta_n} - \frac{\alpha_n}{\beta_n}\cdot f'(0) \right) \right|$$

$$\left| \frac{f(\beta_n) - f(0)}{\beta_n - \alpha_n} \right| = \left| \frac{\beta_n}{\beta_n - \alpha_n} \cdot \frac{f(\beta_n) - f(0)}{\beta_n - 0} \right| \le M \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} \right| \to M \cdot |f'(0)|.$$

「Section 5.22. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable s.t $f'(t) \neq 1$ for all $t \in \mathbb{R}$.

1). f has at most one fixed point.

2). $f(t) = t + (1+e^t)^{-1}$ has no fixed point, although $0 < f'(t) < 1$ for all $t \in \mathbb{R}$.

3). If $\sup_{t \in \mathbb{R}} |f'(t)| < 1$, then f has a fixed point.

Proof. Let α, β be two distinct fixed point. Then, by the Mean-value Theorem,

$$\frac{f(\beta) - f(\alpha)}{\beta - \alpha} = 1 = f'(c) \quad \text{for some } c.$$

It is Contradiction. 2) is trivial. $f(t) - t = \frac{1}{1+e^t} \neq 0$. But

$$f'(t) = 1 - (1+e^t)^{-2} \cdot e^t = 1 - \frac{e^t}{(1+e^t)^2}$$

is contained in $(0,1)$. Let's show 3), The Banach-fixed Point Theorem.

Solve in Next page

Problem. Let $f: \mathbb{R} \to \mathbb{R}$ be a differentiable with $\sup|f'(x)| < 1$.
Then f has a fixed point.

Proof. Construct a sequence. Let $x_1 = \alpha \in \mathbb{R}$, $x_{n+1} = f(x_n)$ $n \geq 1$.
Take $\beta = \sup|f'(x)| < 1$. Given $n \in \mathbb{N}$, if $x_n = x_{n-1}$ then x_{n-1} is the fixed point. If $x_n = x_{n-1}$,

$$|x_{n+1} - x_n| = |f(x_n) - f(x_{n-1})| = |x_n - x_{n-1}| \cdot |f'(c)| \leq \beta \cdot |x_n - x_{n-1}| \quad \text{by the MVT.}$$

Now,
$$\begin{aligned} |x_{n+1} - x_n| &= |f(x_n) - f(x_{n-1})| \leq \beta |x_n - x_{n-1}| \\ &= \beta \cdot |f(x_{n-1}) - f(x_{n-2})| \leq \beta^2 |x_{n-1} - x_{n-2}| \\ &\;\vdots \\ &\leq \beta^{n-1} |x_2 - x_1|. \end{aligned}$$

Given $\varepsilon > 0$, put $N \in \mathbb{N}$ such that $\beta^{N-1}|x_2 - x_1| < \varepsilon(1-\beta)$.
Then for any $m \geq n \geq N$,

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \cdots + |x_{n+1} - x_n| \\ &\leq (\beta^{m-2} + \beta^{m-1} + \cdots + \beta^{n-1}) \cdot |x_2 - x_1| \\ &= \beta^{n-1}(1 + \beta + \cdots + \beta^{m-n-1}) \cdot |x_2 - x_1| \\ &< \beta^{n-1}\left(\sum_{k=0}^{\infty} \beta^n\right) \cdot |x_2 - x_1| = \beta^{n-1} \cdot \frac{1}{1-\beta} \cdot |x_2 - x_1| < \varepsilon. \end{aligned}$$

Therefore $\{x_n\}$ is cauchy sequence in $\mathbb{R}$, the complete metric space, thus

$$x^* = \lim_{n \to \infty} x_n$$

exists. Finally, the differentiable map continuous,

$$f(x^*) = f\left(\lim_{n \to \infty} x_n\right) = \lim_{n \to \infty} f(x_n) = \lim_{n \to \infty} x_{n-1} = x^*. \quad \blacksquare$$

Note: 수축사상 → MVT → Difference of Adjacent 가 지수적으로 감소 → 코시수열 → 완비수렴 → 연속수렴보존

「If a Continuous map $f : \mathbb{R} \to \mathbb{R}$ satisfies : f has a local minimum at for any $x \in \mathbb{R}$, then f is Constant.

Proof. Let $a \in \mathbb{R}$ be given. Consider a set

$$A \stackrel{\text{def}}{=} \{x \in \mathbb{R} \mid f(a) \le f(x)\} = f^{-1}\left[[f(a), \infty)\right]$$

Since f is Continuous, A is closed set. Meanwhile, for any $b \in A$, b is a local minimum:

there exists an open $\mathcal{U}$ containing b s.t. $\forall x \in \mathcal{U}$, $f(b) \le f(x)$

But $f(a) \le f(b)$, thus $\mathcal{U} \subseteq A$. Now, A is non-empty clopen, thus $A = \mathbb{R}$.

Finally, for any $a, b \in \mathbb{R}$, $f(a) \le f(b)$ and $f(a) \ge f(b)$

「Section 5.23.

Define $f(x) = \frac{x^3+1}{3}$. It has three fixed point α, β, γ where
$-2 < \alpha < -1$, $0 < \beta < 1$, $1 < \gamma < 2$

For arbitrary $x_1 \in \mathbb{R}$, define $\{x_n\}$ as $x_{n+1} = f(x_n)$, $n \geq 1$.

1). If $x_1 < \alpha$, then $x_n \to -\infty$ as $n \to \infty$

2) If $\alpha < x_1 < \gamma$, then $x_n \to \beta$ as $n \to \infty$

3) If $\gamma < x_1$, then $x_n \to \infty$ as $n \to \infty$

Proof. observe that $f'(x) = x^2$. And, for each $n \geq 1$,

$$x_{n+1} - x_n = f(x_n) - x_n = \frac{x_n^3+1}{3} - x_n$$

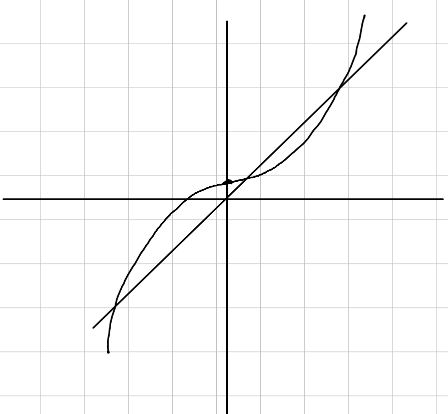
Suppose that $x_1 < \alpha$. For $n=1$, $x_2 - x_1 = f(x_1) - x_1 < 0$.
And, if $x_n - x_{n-1} < 0$, then for some t_n between x_n and x_{n-1},

$$x_{n+1} - x_n = f(x_n) - f(x_{n-1}) = f'(t_n)(x_n - x_{n-1}) < 0 \quad \text{by } f'(t_n) > 0.$$

Inductively, $\{x_n\}$ is decreasing. Moreover,

$$x_{n+1} - x_n = f(x_n) - f(x_{n-1}) = f'(t_n)(x_n - x_{n-1}) \leq x_n - x_{n-1} \quad \text{by } f'(t_n) > 1.$$

Therefore, $x_n - x_1 = \sum_{k=1}^{n-1} x_{k+1} - x_k \leq (n-1)(x_2 - x_1) \to -\infty$ as $n \to \infty$,
thus $x_n \to -\infty$. Similarly, $x_n \to \infty$ if $x_1 > \gamma$.



To. 정종원

행복하세요

2026. 05. 22 금
시립대 세미나실에서